

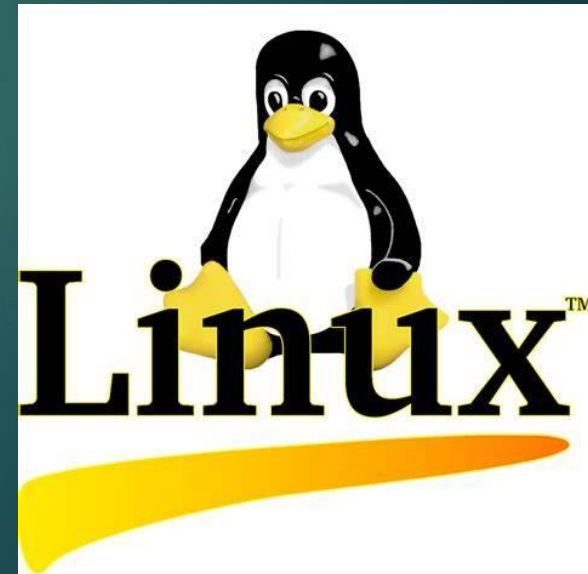


Module 8: Linux Basics and Working in the Linux Shell

- ENDPOINT SECURITY | CISCO NETWORKING ACADEMY
- PREPARED BY: KUDZAISHE MAJEZA

8.0 Introduction to Linux

- ▶ Linux is an open-source operating system.
 - ▶ Open-source means the source code is publicly available for modification.
 - ▶ It is widely used in servers, cybersecurity, networking, and cloud computing.
 - ▶ Linux is known for stability, security, and performance.



8.1 Linux Basics

- ▶ Linux is multi-user (multiple users can use the system).
 - ▶ Linux is multitasking (can run multiple programs at once).
 - ▶ The kernel is the core part of the operating system.
 - ▶ Distributions (distros) are different versions of Linux like Ubuntu and Kali Linux.

8.2.1 The Linux Shell

- ▶ The Shell is a Command-Line Interface (CLI).
 - ▶ CLI means Command-Line Interface.
 - ▶ It allows users to interact with the system by typing commands.
 - ▶ Bash (Bourne Again Shell) is the most common shell.

Why Use the Linux Shell?

- ▶ Provides direct control over the operating system.
 - ▶ Faster and more powerful for administrators.
 - ▶ Essential for cybersecurity and system management.
 - ▶ Supports automation using shell scripts.

8.2.2 Basic Linux Commands

- ▶ pwd – Print Working Directory (shows current location).
 - ▶ ls – List directory contents.
 - ▶ whoami – Displays current logged-in user.
 - ▶ clear – Clears the terminal screen.

8.2.3 File and Directory Commands

- ▶ `cd` – Change Directory.
- ▶ `mkdir` – Make Directory (create folder).
- ▶ `rmdir` – Remove empty directory.
- ▶ `rm` – Remove files or directories.
- ▶ `cp` – Copy files.
- ▶ `mv` – Move or rename files.

8.2.4 Working with Text Files

- ▶ cat – Display file content.
 - ▶ nano – Terminal-based text editor.
 - ▶ grep – Search for specific text in files.
 - ▶ touch – Create a new empty file.

8.2.5 Importance of Text Files in Linux

- ▶ System configurations are stored in text files.
 - ▶ Logs are stored as readable text files.
 - ▶ Scripts are written and executed as text files.
 - ▶ Text files make troubleshooting easier.

Module 8 Summary

- ▶ Linux is a powerful and secure open-source operating system.
 - ▶ The Shell allows command-based interaction with the system.
 - ▶ Commands help manage files and directories efficiently.
 - ▶ Text files are central to Linux configuration and management.